BABU BANARASI DAS UNIVERSITY

2026 – 2027



SCHOOL OF COMPUTER APPLICATION

No SQL

ASSIGNMENT 2: - Students Management System Using MongoDB

SUBMITTED BY: -

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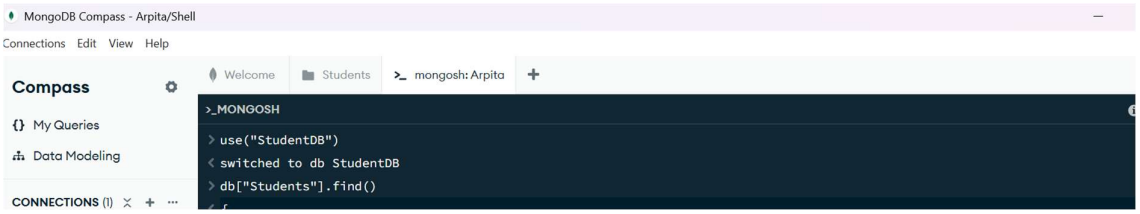
Mr. Harendra Singh

1.CREATE DATABASE

MongoDB Query:

Use Students

Explanation: Creates or switches to the Student database.

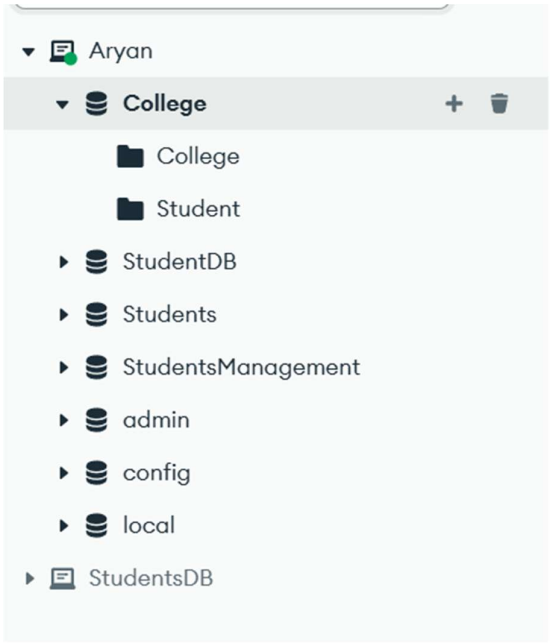


2.CREATE COLLEECTION

MongoDB Query:

db. Create Collection("students")

Explanation: Creates a collection named Student.



3. INSERT 50 STUDENTS

MongoDB Query:

db. students. Insert Many([])

Explanation: Insert multiple student documents into the collection.

{"rollno": 1, "name": "Aarav Sharma", "age": 18, "marks": 85, "city": "Delhi"},

{"rollno": 2, "name": "Priya Verma", "age": 19, "marks": 78, "city": "Lucknow"},

{"rollno": 3, "name": "Rohan Gupta", "age": 20, "marks": 92, "city": "Jaipur"},

{"rollno": 4, "name": "Sneha Singh", "age": 18, "marks": 67, "city": "Patna"},

{"rollno": 5, "name": "Aditya Mishra", "age": 21, "marks": 74, "city": "Varanasi"},

{"rollno": 6, "name": "Neha Yadav", "age": 19, "marks": 88, "city": "Kanpur"},

{"rollno": 7, "name": "Vikram Joshi", "age": 22, "marks": 59, "city": "Dehradun"},

{"rollno": 8, "name": "Anjali Pandey", "age": 18, "marks": 91, "city": "Prayagraj"},

{"rollno": 9, "name": "Karan Mehta", "age": 20, "marks": 63, "city": "Mumbai"},

{"rollno": 10, "name": "Pooja Kumari", "age": 19, "marks": 82, "city": "Ranchi"},

{"rollno": 11, "name": "Rahul Tiwari", "age": 21, "marks": 70, "city": "Bhopal"},

{"rollno": 12, "name": "Simran Kaur", "age": 18, "marks": 95, "city": "Amritsar"},

{"rollno": 13, "name": "Amit Chauhan", "age": 20, "marks": 56, "city": "Agra"},

{"rollno": 14, "name": "Kavya Nair", "age": 19, "marks": 89, "city": "Kochi"},

{"rollno": 15, "name": "Deepak Rawat", "age": 22, "marks": 64, "city": "Haridwar"},

{"rollno": 16, "name": "Ritu Agarwal", "age": 18, "marks": 76, "city": "Indore"},

{"rollno": 17, "name": "Manish Dubey", "age": 20, "marks": 81, "city": "Gorakhpur"},

{"rollno": 18, "name": "Ayesha Khan", "age": 19, "marks": 87, "city": "Hyderabad"},

{"rollno": 19, "name": "Suresh Patel", "age": 21, "marks": 52, "city": "Ahmedabad"},

{"rollno": 20, "name": "Divya Saxena", "age": 18, "marks": 93, "city": "Bareilly"},

{"rollno": 21, "name": "Nikhil Bansal", "age": 20, "marks": 69, "city": "Meerut"},

{"rollno": 22, "name": "Meena Kumari", "age": 19, "marks": 73, "city": "Gaya"},

{"rollno": 23, "name": "Harsh Vardhan", "age": 22, "marks": 84, "city": "Chandigarh"},

{"rollno": 24, "name": "Isha Srivastava", "age": 18, "marks": 90, "city": "Lucknow"},

{"rollno": 25, "name": "Gaurav Thakur", "age": 21, "marks": 61, "city": "Shimla"},
{"rollno": 26, "name": "Tanvi Deshmukh", "age": 19, "marks": 79, "city": "Pune"},
{"rollno": 27, "name": "Sandeep Yadav", "age": 20, "marks": 58, "city": "Azamgarh"},
{"rollno": 28, "name": "Nisha Rathore", "age": 18, "marks": 86, "city": "Jodhpur"},
{"rollno": 29, "name": "Ajay Kushwaha", "age": 22, "marks": 66, "city": "Mirzapur"},
{"rollno": 30, "name": "Shruti Bhatt", "age": 19, "marks": 94, "city": "Nainital"},
{"rollno": 31, "name": "Pankaj Sinha", "age": 21, "marks": 72, "city": "Muzaffarpur"},
{"rollno": 32, "name": "Ritika Jain", "age": 18, "marks": 83, "city": "Surat"},
{"rollno": 33, "name": "Mohit Kapoor", "age": 20, "marks": 55, "city": "Gurugram"},
{"rollno": 34, "name": "Swati Choudhary", "age": 19, "marks": 77, "city": "Noida"},
{"rollno": 35, "name": "Arjun Reddy", "age": 21, "marks": 96, "city": "Hyderabad"},
{"rollno": 36, "name": "Komal Bisht", "age": 18, "marks": 68, "city": "Almora"},
{"rollno": 37, "name": "Sachin Tripathi", "age": 22, "marks": 62, "city": "Ayodhya"},
{"rollno": 38, "name": "Rekha Devi", "age": 19, "marks": 75, "city": "Darbhanga"},
{"rollno": 39, "name": "Yash Malhotra", "age": 20, "marks": 88, "city": "Delhi"},
{"rollno": 40, "name": "Fatima Siddiqui", "age": 18, "marks": 80, "city": "Aligarh"},
{"rollno": 41, "name": "Rajesh Maurya", "age": 21, "marks": 54, "city": "Jaunpur"},
{"rollno": 42, "name": "Anushka Das", "age": 19, "marks": 91, "city": "Kolkata"},
{"rollno": 43, "name": "Vivek Pathak", "age": 20, "marks": 71, "city": "Rewa"},
{"rollno": 44, "name": "Payal Goyal", "age": 18, "marks": 85, "city": "Mathura"},
{"rollno": 45, "name": "Lokesh Meena", "age": 22, "marks": 60, "city": "Kota"},
{"rollno": 46, "name": "Bhavna Solanki", "age": 19, "marks": 74, "city": "Udaipur"},
{"rollno": 47, "name": "Tarun Bhardwaj", "age": 20, "marks": 97, "city": "Rohtak"},
{"rollno": 48, "name": "Jyoti Kashyap", "age": 18, "marks": 65, "city": "Ambala"},
{"rollno": 49, "name": "Zubair Ansari", "age": 21, "marks": 78, "city": "Bhopal"},
{"rollno": 50, "name": "Mansi Kulkarni", "age": 19, "marks": 89, "city": "Nagpur"}

```
>_MONGOSH
College> db.students.insertMany([
    {"rollno": 1, "name": "Aarav Sharma", "age": 18, "marks": 85, "city": "Delhi"},
    {"rollno": 2, "name": "Priya Verma", "age": 19, "marks": 78, "city": "Lucknow"},
    {"rollno": 3, "name": "Rohan Gupta", "age": 20, "marks": 92, "city": "Jaipur"},
    {"rollno": 4, "name": "Sneha Singh", "age": 18, "marks": 67, "city": "Patna"},
    {"rollno": 5, "name": "Aditya Mishra", "age": 21, "marks": 74, "city": "Varanasi"},
    {"rollno": 6, "name": "Neha Yadav", "age": 19, "marks": 88, "city": "Kanpur"},
    {"rollno": 7, "name": "Vikram Joshi", "age": 22, "marks": 59, "city": "Dehradun"},
    {"rollno": 8, "name": "Anjali Pandey", "age": 18, "marks": 91, "city": "Prayagraj"},
    {"rollno": 9, "name": "Karan Mehta", "age": 20, "marks": 63, "city": "Mumbai"},
    {"rollno": 10, "name": "Pooja Kumari", "age": 19, "marks": 82, "city": "Ranchi"},
    {"rollno": 11, "name": "Rahul Tiwari", "age": 21, "marks": 70, "city": "Bhopal"},
    {"rollno": 12, "name": "Simran Kaur", "age": 18, "marks": 95, "city": "Amritsar"},
    {"rollno": 13, "name": "Amit Chauhan", "age": 20, "marks": 56, "city": "Agra"},
    {"rollno": 14, "name": "Kavya Nair", "age": 19, "marks": 89, "city": "Kochi"},
    {"rollno": 15, "name": "Deepak Rawat", "age": 22, "marks": 64, "city": "Haridwar"},
    {"rollno": 16, "name": "Ritu Agarwal", "age": 18, "marks": 76, "city": "Indore"},
    {"rollno": 17, "name": "Manish Dubey", "age": 20, "marks": 81, "city": "Gorakhpur"},
    {"rollno": 18, "name": "Ayesha Khan", "age": 19, "marks": 87, "city": "Hyderabad"},
    {"rollno": 19, "name": "Suresh Patel", "age": 21, "marks": 52, "city": "Ahmedabad"},
    {"rollno": 20, "name": "Divya Saxena", "age": 18, "marks": 93, "city": "Bareilly"},
    {"rollno": 21, "name": "Nikhil Bansal", "age": 20, "marks": 69, "city": "Meerut"},
```

4. DISPLAY ALL STUDENTS

MongoDB Query:

db. students. Find()

Explanation: Retrieves documents from the students collection.

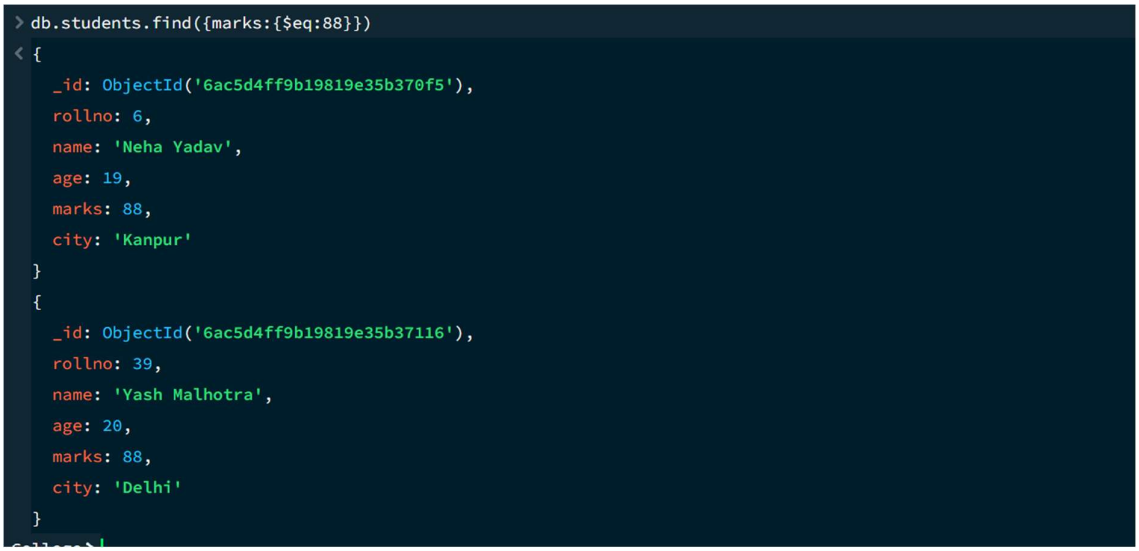
```
> db.students.find()
< {
    _id: ObjectId('6ac5d4ff9b19819e35b370f0'),
    rollno: 1,
    name: 'Aarav Sharma',
    age: 18,
    marks: 85,
    city: 'Delhi'
  }
  {
    _id: ObjectId('6ac5d4ff9b19819e35b370f1'),
    rollno: 2,
    name: 'Priya Verma',
    age: 19,
    marks: 78,
    city: 'Lucknow'
  }
```

5. EQUAL TO - $eq

MongoDB Query:

db. students. Find ({marks: 88})

Explanation: Finds students whose marks are exactly 88.



6. GREATER THAN - $gt

MongoDB Query:

db.students.find({marks:{$gt:85}})

Explanation: Find students whose marks are greater than 85

```
>_MONGOSH
> db.students.find({marks:{$gt:88}})
< {
    _id: ObjectId('6ac5d4ff9b19819e35b370f2'),
    rollno: 3,
    name: 'Rohan Gupta',
    age: 20,
    marks: 92,
    city: 'Jaipur'
  }
  {
    _id: ObjectId('6ac5d4ff9b19819e35b370f7'),
    rollno: 8,
    name: 'Anjali Pandey',
    age: 18,
    marks: 91,
    city: 'Prayagraj'
  }
  {
    _id: ObjectId('6ac5d4ff9b19819e35b370fb'),
    rollno: 12,
    name: 'Simran Kaur',
    age: 18,
```

7. LESS THAN - $lt

MongoDB Query:

db. students.find({marks:{$lt:60}})

Explanation: Finds students whose marks are less than 60.

```
>_MONGOSH
> db.students.find({marks:{$lt:88}})
< {
    _id: ObjectId('6ac5d4ff9b19819e35b370f0'),
    rollno: 1,
    name: 'Aarav Sharma',
    age: 18,
    marks: 85,
    city: 'Delhi'
  }
  {
    _id: ObjectId('6ac5d4ff9b19819e35b370f1'),
    rollno: 2,
    name: 'Priya Verma',
    age: 19,
    marks: 78,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6ac5d4ff9b19819e35b370f3'),
    rollno: 4,
    name: 'Sneha Singh',
    age: 18,
```

8. GREATER THAN OR EQUAL TO - $gte

MongoDB Query:

db.students.finds({marks:{$gte:90}})

Explanation: Finds students whose marks are 90 or above.

```
> db.students.find({marks:{$gte:88}})
< {
    _id: ObjectId('6ac5d4ff9b19819e35b370f2'),
    rollno: 3,
    name: 'Rohan Gupta',
    age: 20,
    marks: 92,
    city: 'Jaipur'
  }
  {
    _id: ObjectId('6ac5d4ff9b19819e35b370f5'),
    rollno: 6,
    name: 'Neha Yadav',
    age: 19,
    marks: 88,
    city: 'Kanpur'
  }
  {
    _id: ObjectId('6ac5d4ff9b19819e35b370f7'),
    rollno: 8,
    name: 'Anjali Pandey',
    age: 18,
```

9. LESS THAN OR EQUAL TO - $lte

MongoDB Query:

db.students.finds({marks:{$lte:70}})

Explanation: Finds students whose marks are 70 or below.

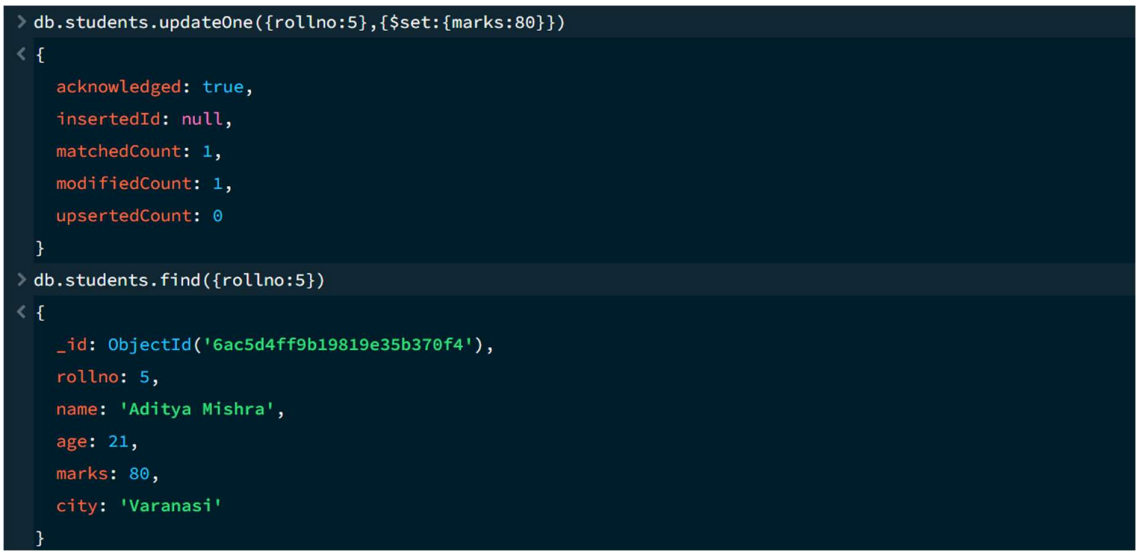
```
> db.students.find({marks:{$lte:75}})
< {
    _id: ObjectId('6ac5d4ff9b19819e35b370f3'),
    rollno: 4,
    name: 'Sneha Singh',
    age: 18,
    marks: 67,
    city: 'Patna'
  }
  {
    _id: ObjectId('6ac5d4ff9b19819e35b370f4'),
    rollno: 5,
    name: 'Aditya Mishra',
    age: 21,
    marks: 74,
    city: 'Varanasi'
  }
```

10. UPDATE OPERATION

MongoDB Query:

db.students.updateOne({roll:5},{$set:{marks:95}})

Explanation: Updates the marks to 95 of the student with roll number 5.



11. DELETE ONE OPERATION

MongoDB Query:

db.students.deleteOne({roll:10})

Explanation: Deletes one document matching roll number 10.



12. DELETE MANY OPERATION

MongoDB Query:

db.students.deleteMany({marks:{$lt:60}})

Explanation: deletes all students having marks below 60.

```
> db.students.deleteMany({marks:{$lt:60}})
< {
    acknowledged: true,
    deletedCount: 6
  }
```

13. AND OPERATION- $and

MongoDB Query:

db.students.find({$and:[{marks:{$gt:80}},{age:19}]})

Explanation: Finds students satisfying both conditions.

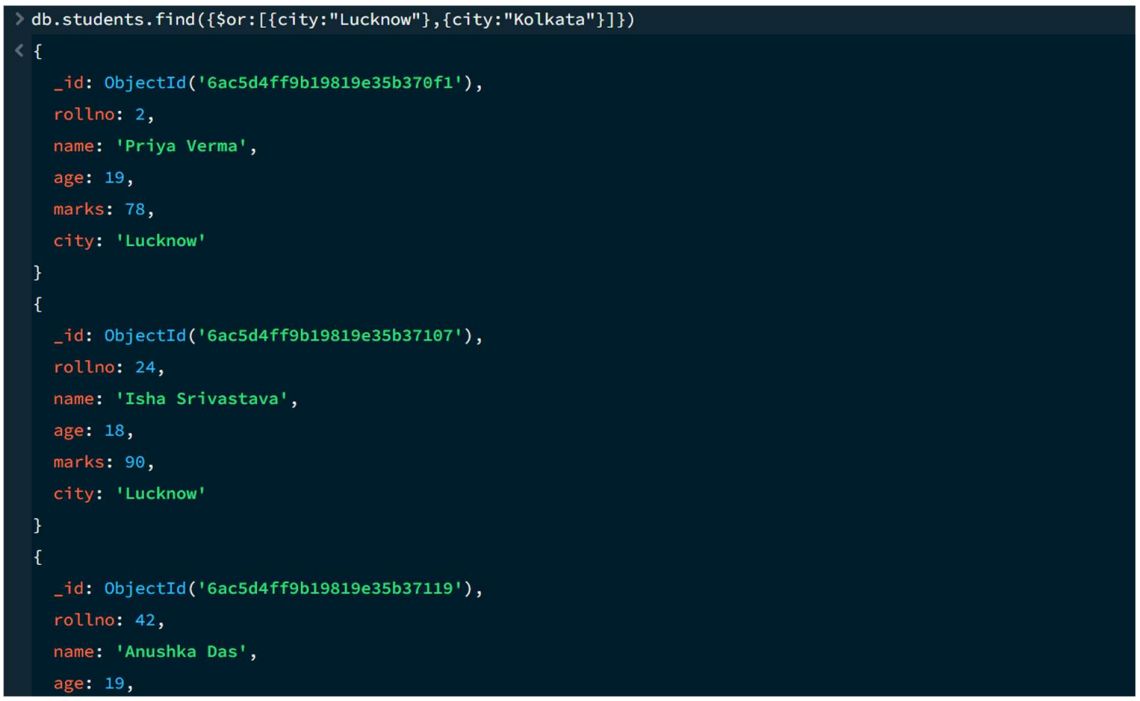
```
> db.students.find({$and:[{marks:{$gt:80}},{age:19}]})
< {
    _id: ObjectId('6ac5d4ff9b19819e35b370f5'),
    rollno: 6,
    name: 'Neha Yadav',
    age: 19,
    marks: 88,
    city: 'Kanpur'
  }
  {
    _id: ObjectId('6ac5d4ff9b19819e35b370fd'),
    rollno: 14,
    name: 'Kavya Nair',
    age: 19,
    marks: 89,
    city: 'Kochi'
  }
```

14: OR OPERATION - $or

MongoDB Query:

db.students.find({$or:[{city:”Lucknow”},{city:”Kanpur”}]})

Explanation: Finds students from Lucknow or Kanpur.



15. SORT OPERATION

A. MongoDB Query:

db.students.find().sort({marks:1})

Explanation: Sorts students by marks from lowest to highest .use 1 for ascending order.

```
> db.students.find().sort({marks:1})
< {
    _id: ObjectId('6ac5d4ff9b19819e35b3711c'),
    rollno: 45,
    name: 'Lokesh Meena',
    age: 22,
    marks: 60,
    city: 'Kota'
  }
  {
    _id: ObjectId('6ac5d4ff9b19819e35b37108'),
    rollno: 25,
    name: 'Gaurav Thakur',
    age: 21,
    marks: 61,
    city: 'Shimla'
  }
  {
    _id: ObjectId('6ac5d4ff9b19819e35b37114'),
    rollno: 37,
    name: 'Sachin Tripathi',
    age: 22,
```

B. MongoDB Query:

db.students.find().sort({marks: -1})

Explanation: Sorts students by marks from lowest to highest .Use -1 for descending order.

```
> db.students.find().sort({marks:-1})
< {
    _id: ObjectId('6ac5d4ff9b19819e35b3711e'),
    rollno: 47,
    name: 'Tarun Bhardwaj',
    age: 20,
    marks: 97,
    city: 'Rohtak'
  }
  {
    _id: ObjectId('6ac5d4ff9b19819e35b37112'),
    rollno: 35,
    name: 'Arjun Reddy',
    age: 21,
    marks: 96,
    city: 'Hyderabad'
  }
  {
    _id: ObjectId('6ac5d4ff9b19819e35b370fb'),
    rollno: 12,
    name: 'Simran Kaur',
    age: 18,
```

16. LIMIT OPERATION

MongoDB Query:

db.students.find().limit(5)

Explanation: Display only the first five documents.

```
> db.students.find().limit(5)
< {
    _id: ObjectId('6ac5d4ff9b19819e35b370f0'),
    rollno: 1,
    name: 'Aarav Sharma',
    age: 18,
    marks: 85,
    city: 'Delhi'
  }
  {
    _id: ObjectId('6ac5d4ff9b19819e35b370f1'),
    rollno: 2,
    name: 'Priya Verma',
    age: 19,
    marks: 78,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6ac5d4ff9b19819e35b370f2'),
    rollno: 3,
    name: 'Rohan Gupta',
    age: 20,
```

17. SEARCH BY CITY

MongoDB Query:

db.students.find({city:"Lucknow"})

Explanation: Finds all students whose city is Lucknow.

```
> db.students.find({city:"Lucknow"})
< {
    _id: ObjectId('6ac5d4ff9b19819e35b370f1'),
    rollno: 2,
    name: 'Priya Verma',
    age: 19,
    marks: 78,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6ac5d4ff9b19819e35b37107'),
    rollno: 24,
    name: 'Isha Srivastava',
    age: 18,
    marks: 90,
    city: 'Lucknow'
  }
```

18. COUNT DOCUMENTS

MongoDB Query:

db.students.countDocuments()

Explanation: Counts Documents currently present in the collection (performing after the delete function).

```
> db.students.countDocuments()
< 43
College>
```